College Champlain Saint-Lambert

GAME DEVELOPMENT WITH THE UNITY CROSS-PLATFORM GAME ENGINE

420-N44-LA section 00002
Teacher: Kawser Nafi

Final Project

Checkpoint 02: (6.25%)

Due date: April 9, 2026

Mustafa Amin

Team Information

1. **The name of the team is:** TeseroRun
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3. **Git repository:** https://github.com/mustafakhafajy10/TeseroRun.git

1. Ten Features / Work Steps of the Project

- Design and implementation of the Main Menu scene.
- Creation of the player character and setup in Unity.
- Development of the player movement system.
- Development of the player attack/combat system.
- Coin collection and score tracking system.
- Key collection and level progression system.
- Enemy AI and enemy combat interactions.
- Health system with health bar integration.
- NPC helper unlock and support behavior.
- Scene management and progression between menu and gameplay scenes.

2. Four Implemented Features / Work Steps

2.1 Player Movement System

The player can move horizontally and jump. This feature forms the foundation of gameplay and allows the player to navigate the environment effectively.

2.2 Coin and Key Collection System

The player can collect coins to increase score and gather keys required for progression. This supports the game objective and introduces reward-based gameplay.

2.3 Enemy Combat System

Enemies are able to interact with the player, attack, and receive damage. This adds challenge and creates the action component of the game.

2.4 Main Menu and Playable Level 1 Scene

The project contains a functional Main Menu scene and a playable Level 1 scene. This allows the current progress of the game to be demonstrated clearly.

Reason for Choosing These Four Features

I chose these four features were selected because they represent the main gameplay loop of the project. They also demonstrate meaningful progress in level design, scene creation, scripting, prefabs, physics, and character interaction. In addition, these features satisfy the main technical requirements of the checkpoint.

3. Four-Week Development Plan

Week 1

Improve Level 1 gameplay and fix current script and scene issues. Refine UI elements such as score, key display, and health bar. Test player controls, collisions, and enemy interactions.

Week 2

Develop and improve Level 2. Add more environment elements, enemy placement, and level design features. Continue improving gameplay balance and player experience.

Week 3

Develop and implement Level 3. Add advanced gameplay elements such as boss-level mechanics or a final challenge system. Test progression from Level 2 to Level 3.

Week 4

Final debugging and overall project polish. Organize project files, scripts, and prefabs. Prepare final documentation, GitHub submission, and presentation.

4. Individual Contribution

This project checkpoint was completed individually. All design, development, scene setup, scripting, testing, and project organization were completed by the student.

Responsibilities Completed Individually

- Player movement and combat system
- Scene setup and level design
- UI elements and score/health systems
- Enemy behavior and gameplay testing
- GitHub project management and checkpoint preparation

5. Required Elements Implemented

Scene – Implemented

A functional Main Menu scene and a playable Level 1 scene have been created in Unity.

Character – Implemented

A player character has been added to the project with sprite setup, animation, health, and combat behavior.

Movement Scripts – Implemented

The player movement system has been implemented, including horizontal movement and jumping. Enemy and NPC movement behaviors are also included.

Physics – Implemented

Unity 2D physics is being used for collisions, gravity, rigidbody movement, triggers, damage interaction, and object collection such as coins and keys.

Prefab – Implemented

Multiple prefabs have been created and used in the project, including the player, enemies, coins, keys, camera, game manager, NPC spawn, spikes, and UI-related objects.

6. Conclusion

In this checkpoint, 10 planned features/work steps were identified and 4 important gameplay systems were implemented successfully. The project now includes a playable game structure with a main menu, a character, movement, collection mechanics, enemy interaction, and a working gameplay level. The next phase of development will focus on expanding the game further, improving quality, and preparing the final version of the project.